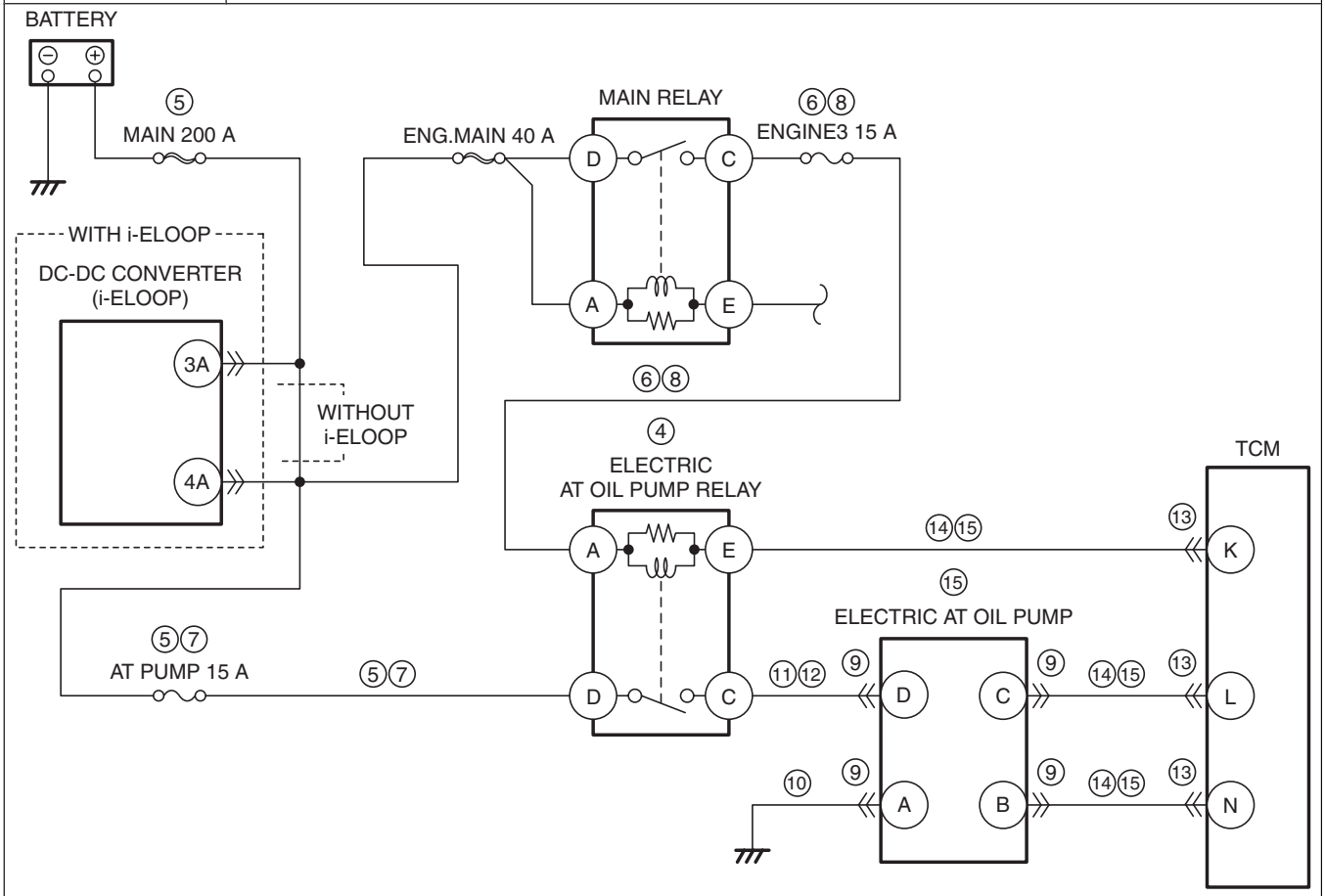
## NO.9 ACCELERATION MALFUNCTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

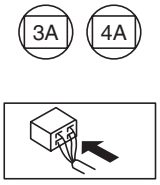
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| 9 | ACCELERATION MALFUNCTION |
|---|---|
| DESCRIPTION | • Acceleration from i-stop is not smooth.<br>• Suppresses (shock) when accelerating vehicle from i-stop.<br>• Engine vibration increases when engine is restarted. |
| POSSIBLE CAUSE | **Note**<br>• MTX vehicles are not included because the driver releases the brake pedal and depresses the clutch pedal to accelerate.<br><br>**ATX system malfunction**<br>• Electric AT oil pump malfunction (operation malfunction, insufficient pressure)<br>— Electric AT oil pump relay malfunction (stuck open)<br>— Short to ground or open circuit in electric AT oil pump relay power supply circuit (without i-ELOOP)<br>• Short to ground in wiring harness between MAIN 200 A fuse and electric AT oil pump relay terminal D<br>• MAIN 200 A fuse and/or AT PUMP 15 A fuse malfunction<br>• Open circuit in wiring harness between battery positive terminal and electric AT oil pump relay terminal D<br>— Short to ground or open circuit in electric AT oil pump relay power supply circuit (without i-ELOOP)<br>• Short to ground in wiring harness between ENGINE3 15 A fuse and electric AT oil pump relay terminal A<br>• ENGINE3 15 A fuse malfunction<br>• Open circuit in wiring harness between main relay terminal C and electric AT oil pump relay terminal A<br>— Short to ground or open circuit in electric AT oil pump relay power supply circuit (with i-ELOOP)<br>• Short to ground in wiring harness between AT PUMP 15 A fuse and electric AT oil pump relay terminal D<br>• AT PUMP 15 A fuse malfunction<br>• Open circuit in wiring harness between DC-DC converter (i-ELOOP) terminal 4A and electric AT oil pump relay terminal D<br>— Short to ground or open circuit in electric AT oil pump relay power supply circuit (with i-ELOOP)<br>• Short to ground in wiring harness between ENGINE3 15 A fuse and electric AT oil pump relay terminal A<br>• ENGINE3 15 A fuse malfunction<br>• Open circuit in wiring harness between main relay terminal C and electric AT oil pump relay terminal A<br>— Electric AT oil pump connector or terminals malfunction<br>— Open circuit in wiring harness between electric AT oil pump terminal A and body ground<br>— Short to ground in wiring harness between electric AT oil pump relay terminal C and electric AT oil pump terminal D<br>— Open circuit in wiring harness between electric AT oil pump relay terminal C and electric AT oil pump terminal D<br>— TCM connector or terminals malfunction<br>— Short to ground in wiring harness between the following terminals:<br>• Electric AT oil pump relay terminal E—TCM terminal K<br>• Electric AT oil pump terminal C—TCM terminal L<br>• Electric AT oil pump terminal B—TCM terminal N<br>— Open circuit in wiring harness between the following terminals:<br>• Electric AT oil pump relay terminal E—TCM terminal K<br>• Electric AT oil pump terminal C—TCM terminal L<br>• Electric AT oil pump terminal B—TCM terminal N<br>— Electric AT oil pump malfunction<br>— TCM malfunction<br>• ATX malfunction<br>**Hill launch assist function system (DSC) malfunction**<br>• False detection of inclination angle (cannot calculate correct road slope)<br>— Low-G (XY) sensor (built-into SAS control module) malfunction (In this case, the SAS control module records DTCs C0061:29 and C0062:29.)<br>— Low-G (XY) sensor (built-into SAS control module) initialization malfunction (In this case, the SAS control module records DTC U2100:00.)<br>• False detection of brake fluid pressure<br>— Brake fluid pressure sensor (built-into DSC HU/CM) malfunction<br>• Cannot maintain brake fluid pressure<br>— DSC HU/CM malfunction |

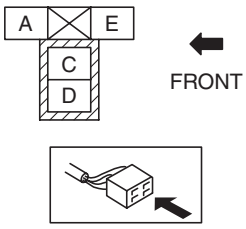
| 9 | ACCELERATION MALFUNCTION |
|---|---|



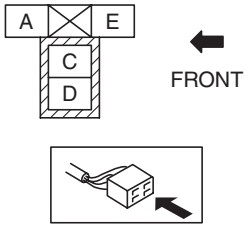

DC-DC CONVERTER (i-ELOOP) WIRING HARNESS-SIDE CONNECTOR



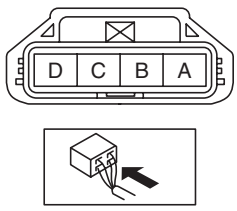

MAIN RELAY (RELAY AND FUSE BLOCK)



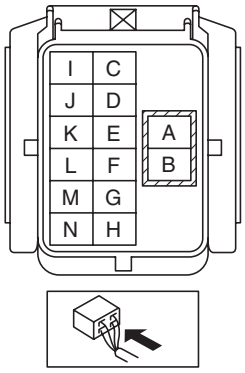

ELECTRIC AT OIL PUMP RELAY (RELAY AND FUSE BLOCK)




ELECTRIC AT OIL PUMP WIRING HARNESS-SIDE CONNECTOR




TCM WIRING HARNESS-SIDE CONNECTOR



**Diagnostic Procedure**

| STEP | INSPECTION | RESULTS | ACTION |
|---|---|---|---|
| 1 | **DETERMINE IF MALFUNCTION CAUSED BY ATX BODY**<br>• Compare the malfunction symptom with the i-stop system stop condition.<br>• Is there any shock or slippage during acceleration with the i-stop system disabled? | Yes | Perform the applicable symptom troubleshooting procedure.<br>(See SYMPTOM TROUBLESHOOTING ITEM TABLE [CW6A-EL].) |
| | | No | Go to the next step. |
| 2 | **VERIFY DTC**<br>• Retrieve the PCM, TCM, DSC HU/CM, SAS control module DTCs using the M-MDS.<br>(See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)<br>(See ON-BOARD DIAGNOSTIC SYSTEM DTC INSPECTION [TCM (CW6A-EL)].)<br>(See DTC INSPECTION [DSC HU/CM].)<br>(See DTC INSPECTION [STANDARD DEPLOYMENT CONTROL SYSTEM].)<br>(See DTC INSPECTION [TWO-STEP DEPLOYMENT CONTROL SYSTEM].)<br>• Are any DTCs present? | Yes | Go to the applicable DTC inspection.<br>(See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)<br>(See ON-BOARD DIAGNOSTIC SYSTEM DTC TABLE [TCM (CW6A-EL)].)<br>(See DTC TABLE [DSC HU/CM].)<br>(See DTC TABLE [STANDARD DEPLOYMENT CONTROL SYSTEM].)<br>(See DTC TABLE [TWO-STEP DEPLOYMENT CONTROL SYSTEM].) |
| | | No | Go to the next step. |
| 3 | **DETERMINE IF MALFUNCTION CAUSE IS BRAKE FLUID PRESSURE SENSOR SIGNAL OR OTHER**<br>• Put the vehicle in an i-stop condition (engine stopped).<br>• Monitor the PCM PID BFP using the M-MDS while the brake is depressed and held with the i-stop function operating.<br>(See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)<br>• Does the monitoring value change? | Yes | Brake fluid pressure sensor (built-into DSC HU/CM) or DSC HU/CM brake pressure hold function malfunction.<br>• Replace the DSC HU/CM.<br>(See DSC HU/CM REMOVAL/INSTALLATION [L.H.D.].)<br>(See DSC HU/CM REMOVAL/INSTALLATION [R.H.D.].) |
| | | No | Go to the next step. |
| 4 | **INSPECT ELECTRIC AT OIL PUMP RELAY**<br>• Switch the ignition off.<br>• Remove the electric AT oil pump relay.<br>(See RELAY LOCATION.)<br>• Inspect the electric AT oil pump relay.<br>(See RELAY INSPECTION.)<br>• Is there any malfunction? | Yes | Replace the electric AT oil pump relay. |
| | | No | Vehicles with i-ELOOP:<br>• Go to Step 7.<br>Vehicles without i-ELOOP:<br>• Go to the next step. |

| STEP | INSPECTION | RESULTS | ACTION |
|---|---|---|---|
| 5 | **INSPECT ELECTRIC AT OIL PUMP RELAY POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT**<br>• Electric AT oil pump relay is removed.<br>• Measure the voltage at the electric AT oil pump relay terminal D (wiring harness-side).<br>• Is the voltage **B+**? | Yes | Go to the next step. |
| | | No | Inspect the MAIN 200 A fuse and AT PUMP 15 A fuse.<br>• If the fuse is blown:<br>— Refer to the wiring diagram and verify whether or not there is a common connector between MAIN 200 A fuse and electric AT oil pump relay terminal D.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has a short to ground.<br>• Replace the malfunctioning fuse.<br>• If the fuse is damaged:<br>— Replace the malfunctioning fuse.<br>• If all fuses are normal:<br>— Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and electric AT oil pump relay terminal D.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has an open circuit. |

| STEP | INSPECTION | RESULTS | ACTION |
|---|---|---|---|
| 6 | **INSPECT ELECTRIC AT OIL PUMP RELAY POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT**<br>• Electric AT oil pump relay is removed.<br>• Switch the ignition ON (engine off).<br>• Measure the voltage at the electric AT oil pump relay terminal A (wiring harness-side).<br>• Is the voltage **B+**? | Yes | Go to Step 9. |
| | | No | Inspect the ENGINE3 15 A fuse.<br>• If the fuse is blown:<br>— Refer to the wiring diagram and verify whether or not there is a common connector between ENGINE3 15 A fuse and electric AT oil pump relay terminal A.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has a short to ground.<br>• Replace the fuse.<br>• If the fuse is damaged:<br>— Replace the fuse.<br>• If the fuse is normal:<br>— Refer to the wiring diagram and verify whether or not there is a common connector between main relay terminal C and electric AT oil pump relay terminal A.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has an open circuit. |

| STEP | INSPECTION | RESULTS | ACTION |
|---|---|---|---|
| 7 | **INSPECT ELECTRIC AT OIL PUMP RELAY POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT**<br>• Electric AT oil pump relay is removed.<br>• Measure the voltage at the electric AT oil pump relay terminal D (wiring harness-side).<br>• Is the voltage **B+**? | Yes | Go to the next step. |
| | | No | Remove the service plug.<br>(See SERVICE PLUG DISCONNECTION/ CONNECTION [i-ELOOP].)<br>Inspect the AT PUMP 15 A fuse.<br>• If the fuse is blown:<br>— Refer to the wiring diagram and verify whether or not there is a common connector between AT PUMP 15 A fuse and electric AT oil pump relay terminal D.<br>**If there is a common connector:**<br>• Determine the malfunctioning location by inspecting the common connector and terminals for corrosion, damage, or disconnection and by inspecting the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness for short to ground.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which is shorted to ground, or the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness.<br>• Replace the fuse.<br>• If the fuse is damaged:<br>— Replace the fuse.<br>• If the fuse is normal:<br>— Refer to the wiring diagram and verify whether or not there is a common connector between DC-DC converter (i-ELOOP) terminal 4A and electric AT oil pump relay terminal D.<br>**If there is a common connector:**<br>• Determine the malfunctioning location by inspecting the common connector and terminals for corrosion, damage, or disconnection and by inspecting the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness for an open circuit.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has an open circuit, or the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness.<br>Reinstall the service plug.<br>(See SERVICE PLUG DISCONNECTION/ CONNECTION [i-ELOOP].) |

| STEP | INSPECTION | RESULTS | ACTION |
|---|---|---|---|
| 8 | **INSPECT ELECTRIC AT OIL PUMP RELAY POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT**<br>• Electric AT oil pump relay is removed.<br>• Switch the ignition ON (engine off).<br>• Measure the voltage at the electric AT oil pump relay terminal A (wiring harness-side).<br>• Is the voltage **B+**? | Yes | Go to the next step. |
| | | No | Inspect the ENGINE3 15 A fuse.<br>• If the fuse is blown:<br>— Refer to the wiring diagram and verify whether or not there is a common connector between ENGINE3 15 A fuse and electric AT oil pump relay terminal A.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has a short to ground.<br>• Replace the fuse.<br>• If the fuse is damaged:<br>— Replace the fuse.<br>• If the fuse is normal:<br>— Refer to the wiring diagram and verify whether or not there is a common connector between main relay terminal C and electric AT oil pump relay terminal A.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has an open circuit. |
| 9 | **INSPECT ELECTRIC AT OIL PUMP CONNECTOR CONDITION**<br>• Switch the ignition off.<br>• Disconnect the electric AT oil pump connector.<br>• Inspect for poor connection (such as damaged/ pulled-out pins, corrosion).<br>• Is there any malfunction? | Yes | Repair or replace the connector and/or terminals. |
| | | No | Go to the next step. |
| 10 | **INSPECT ELECTRIC AT OIL PUMP GROUND CIRCUIT FOR OPEN CIRCUIT**<br>• Electric AT oil pump relay is removed.<br>• Verify that the electric AT oil pump connector is disconnected.<br>• Inspect for continuity between electric AT oil pump terminal A (wiring harness-side) and body ground.<br>• Is there continuity? | Yes | Go to the next step. |
| | | No | Refer to the wiring diagram and verify whether or not there is a common connector between electric AT oil pump terminal A and body ground.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Inspect for the following:<br>— Open circuit between electric AT oil pump and body ground<br>— Loose or lifting ground point<br>• Repair or replace the malfunctioning part. |

| STEP | INSPECTION | RESULTS | ACTION |
|---|---|---|---|
| 11 | **INSPECT ELECTRIC AT OIL PUMP POWER SUPPLY CIRCUIT FOR SHORT TO GROUND**<br>• Electric AT oil pump relay is removed.<br>• Verify that the electric AT oil pump connector is disconnected.<br>• Inspect for continuity between electric AT oil pump relay terminal C (wiring harness-side) and body ground.<br>• Is there continuity? | Yes | Refer to the wiring diagram and verify whether or not there is a common connector between electric AT oil pump relay terminal C and electric AT oil pump terminal D.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has a short to ground. |
| | | No | Go to the next step. |
| 12 | **INSPECT ELECTRIC AT OIL PUMP POWER SUPPLY CIRCUIT FOR OPEN CIRCUIT**<br>• Electric AT oil pump relay is removed.<br>• Verify that the electric AT oil pump connector is disconnected.<br>• Inspect for continuity between electric AT oil pump relay terminal C (wiring harness-side) and electric AT oil pump terminal D (wiring harness-side).<br>• Is there continuity? | Yes | Go to the next step. |
| | | No | Refer to the wiring diagram and verify whether or not there is a common connector between electric AT oil pump relay terminal C and electric AT oil pump terminal D.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has an open circuit. |
| 13 | **INSPECT TCM CONNECTOR CONDITION**<br>• Disconnect the TCM connector.<br>• Inspect for poor connection (such as damaged/pulled-out pins, corrosion).<br>• Is there any malfunction? | Yes | Repair or replace the connector and/or terminals. |
| | | No | Go to the next step. |
| 14 | **INSPECT TCM CIRCUIT FOR SHORT TO GROUND**<br>• Electric AT oil pump relay is removed.<br>• Verify that the electric AT oil pump and TCM connectors are disconnected.<br>• Inspect for continuity between the following terminals (wiring harness-side) and body ground:<br>— Electric AT oil pump relay terminal E<br>— Electric AT oil pump terminal C<br>— Electric AT oil pump terminal B<br>• Is there continuity? | Yes | Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals:<br>• Electric AT oil pump relay terminal E—TCM terminal K<br>• Electric AT oil pump terminal C—TCM terminal L<br>• Electric AT oil pump terminal B—TCM terminal N<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has a short to ground. |
| | | No | Go to the next step. |

| STEP | INSPECTION | RESULTS | ACTION |
|---|---|---|---|
| 15 | **INSPECT TCM CIRCUIT FOR OPEN CIRCUIT**<br>• Electric AT oil pump relay is removed.<br>• Verify that the electric AT oil pump and TCM connectors are disconnected.<br>• Inspect for continuity between the following terminals (wiring harness-side):<br>— Electric AT oil pump relay terminal E—TCM terminal K<br>— Electric AT oil pump terminal C—TCM terminal L<br>— Electric AT oil pump terminal B—TCM terminal N<br>• Is there continuity? | Yes | Replace the electric AT oil pump.<br>(See ELECTRIC AT OIL PUMP REMOVAL/ INSTALLATION [CW6A-EL].) |
| | | No | Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals:<br>• Electric AT oil pump relay terminal E—TCM terminal K<br>• Electric AT oil pump terminal C—TCM terminal L<br>• Electric AT oil pump terminal B—TCM terminal N<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has an open circuit. |
| 16 | Verify the test results.<br>• If normal, return to the diagnostic index to service any additional symptoms.<br>(See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)<br>• If a malfunction remains, inspect the related Service Information and perform the repair or diagnosis.<br>— If the vehicle is repaired, troubleshooting is completed.<br>— If the vehicle is not repaired or additional diagnostic information is not available, replace the PCM.<br>(See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) | | |